



Quran Recitation AI Evaluation System — Client Proposal

Dear Sir, Below is the full plan, cost, and duration for building your Qur'an Recitation Evaluation System.

1. Project Summary

I will develop an AI system that:

Listens to the student's Qur'an recitation

Aligns it with the Quran text (Forced Alignment)

Compares it with expert Sheikh recordings

Detects mistakes in:

words

letters

harakat

tajweed rules

Provides written and audio feedback

Allows the student to retry and improves accuracy

2. Full Workflow

Student records audio

System aligns audio to text (Forced Alignment)

Phoneme confidence is calculated

User audio is compared with Sheikh audio (Reference Matching)

Tajweed errors are detected via timing & pitch analysis

- Errors are detected
- Feedback is generated
- User retries
- System measures improvement

3. Deliverables

- Forced Alignment module (Wav2Vec2)
- Phoneme Confidence Scoring module
- Reference Audio Matching module (DTW)
- Tajweed Analysis module (Pitch/Duration)
- Error classification system
- Feedback generator (text + TTS)
- Retry logic
- JSON API for backend
- Documentation
- Sample audio + sample JSON output

4. Timeline

Phase	Duration
Audio + Data Prep	1 week
Forced Alignment	2 weeks
Error Detection	1 week
Reference Matching	2 weeks
Tajweed Analysis	2 weeks

Phase	Duration
Feedback + Retry	1 week
Testing + Delivery	1 week

Total Duration: 8 Weeks

5. Cost (Junior Engineer Pricing)

Item	Cost (USD)
AI System Development	\$900
Testing & QA	\$100
GPU Compute	\$50–100
Total Estimated Cost	\$1,050 – \$1,200

6. What I Need From You

- Quran text database (already available)
- 4 Sheikh audio datasets (already available)
- Backend endpoint for integration
- Deployment environment details

7. Notes

- The system detects basic tajweed, not advanced scholarly fiqh rules
- AI-based evaluation, suitable for kids and adults
- Backend will be handled by your team, I provide AI only
- System integrates easily with mobile or web apps